

2.1.2 $A \cap B$ ✓

b $A \cup B$ ✓

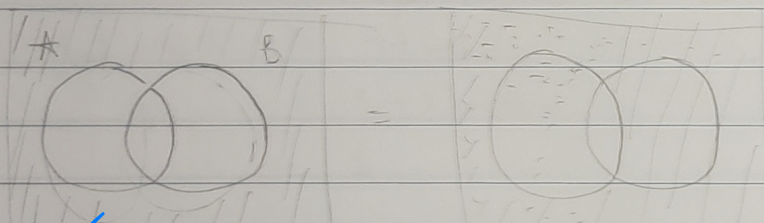
c $\overline{A \cup B}$ ✓

d. $(A \setminus B) \cup (B \setminus A)$ ✓

GOOD!

Section 2.1

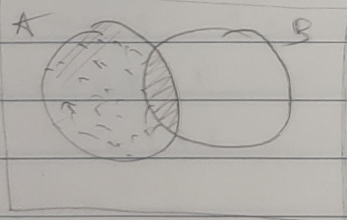
3. $\overline{A \cup B}$



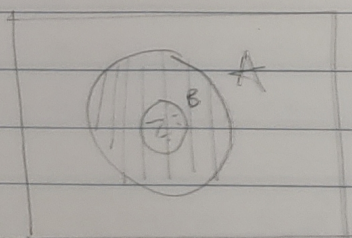
Also need to show that $\overline{(A \cap B)} = \bar{A} \cup \bar{B}$

$\boxed{-\frac{1}{2}}$

a.



b.



2.5g $A = A \cap (B \cup \bar{B}) = A \cap S = A \rightarrow$ trying to show $A = (A \cap B) \cup (A \cap \bar{B})$
(see my proof below)

$$\begin{aligned} (A \cap B) \cup (A \cap \bar{B}) &= A \cap (B \cup \bar{B}) && \text{(distrib. law)} \\ &= A \cap S && (S = B \cup \bar{B}) \\ &= A && (A = A \cap S) \end{aligned}$$

Section 2.4

7. $S = \{A+, A-, B+, B-, AB+, AB-, O+, O-\}$ ✓

11 a.) $2x + .4 + 2y + y = 1$

$.3 + .4 + 3y = 1$

$y = .1$

$\therefore E+ = .2; E- = .1$ ✓

$\boxed{-\frac{1}{2}}$

b.) $P(E_1) = 3P(E_2) = .3 \Rightarrow P(E_1) = .3$ and $P(E_2) = .1$.

We know $\sum_{i=1}^5 P(E_i) = 1$, so $P(E_3) + P(E_4) + P(E_5) = 1 - (.3 + .1) = .6$.

Since E_3, E_4 , and E_5 are equally probable,
 $P(E_3) = P(E_4) = P(E_5) = .2$.

14a. .58 ✓

b. .14 ✓

c. .46 ✓

Going forward, show me the numbers you added to get your final answer.

18a. $S = \{HH, HT, TH, TT\}$ ✓

b. $P(\text{sample point}) = \frac{1}{4}$ since the coin is fair ✓

c. $A \cap B = \{HT, TH\}$

→ this question wants the sample points in A and the sample points in B

d. $P(A) = \frac{2}{4} = \frac{1}{2}$ ✓

$P(B) = \frac{3}{4}$ ✓

$P(A \cap B) = \frac{2}{4} = \frac{1}{2}$ ✓

$P(A \cup B) = \frac{3}{4}$ ✓

$P(\bar{A} \cup B) = 1$ ✓

$A = \{TH, HT\}$

$B = \{TH, HT, HH\}$